

Ian Carlo Carmona Serrano

AI Engineer | LLM Agents in Production

Mexico City, Mexico | iancarlocs@gmail.com | +52 55 7852 0955
linkedin.com/in/engineeria-ian-carmona | github.com/IanCarmona

PROFESSIONAL SUMMARY

AI Engineer with conversational agents running in production and paying customers. I built and operate a LangGraph agent on the WhatsApp Business Platform that handles scheduling, payments, and business queries autonomously. Previously implemented RAG for a conversational system in the financial sector. In parallel, I design Medallion-architecture pipelines in Databricks for a pension fund manager. My focus: reliable, low-cost LLM systems — evals, hallucination control, and cost-per-token optimization.

EXPERIENCE

AI Engineer

November 2025 – Present

Amyra, Mexico City — amyra.com.mx

- Built a conversational agent from scratch on LangGraph and Claude, integrated with the WhatsApp Business Platform, handling scheduling, reminders, membership sales, and business queries without human intervention. Live in production with paying subscription customers.
- Cut cost per message by 70%, from \$1 USD per 10 messages to \$1 USD per 30, through model migration, prompt caching, context pruning, and prompt redesign. On cache-warm turns, cost drops to \$1 per 60 messages.
- Implemented an eval suite over real conversations measuring groundedness and run-to-run consistency, catching hallucinations before they reached end users.
- Migrated the database to Cloud SQL on GCP and own the full stack: frontend, backend, APIs, and infrastructure. Define the technical roadmap and product decisions.

Data Engineer

July 2025 – Present

Profuturo, Mexico City

- Automated the generation and validation of COSAR regulatory filings, eliminating 4+ hours of daily manual processing and removing human error risk from deliverables with regulatory exposure.
- Build Medallion-architecture pipelines in Databricks with multi-cloud ingestion, as part of the FinOps initiative that gave leadership visibility into technology spend via Power BI.
- Design dimensional models with an emphasis on lineage, governance, and data quality.
- Automate internal reporting processes, reducing delivery times and operational friction across teams.

AI Engineer

January 2024 – October 2024

Romboworks, Mexico City

- Developed a conversational system for the financial sector integrating LLMs and NLP techniques, contributing to the functional architecture of the product.
- Implemented RAG with a vector database to optimize context retrieval, increasing model response accuracy by 30%.
- Built the frontend and business logic with React and Node.js, iterating with the NLP team on response quality.

EDUCATION

B.Eng. in Artificial Intelligence

Coursework completed | Thesis in progress

Instituto Politécnico Nacional (IPN) – ESCOM, Mexico City

Technical Degree, Electrical Control Systems

Graduated: June 2021

Instituto Politécnico Nacional (IPN) – CECyT 3, Mexico City

TECHNICAL SKILLS

- **Applied AI:** LangGraph, LangChain, Claude, OpenAI, RAG, vector databases, prompt engineering, prompt caching, evals, agents in production, NLP, machine learning, deep learning
- **Data & Analytics:** Databricks, PySpark, ETL/ELT, Medallion architecture, dimensional modeling, Pandas, NumPy, Power BI
- **Languages & Frameworks:** Python, SQL, JavaScript/TypeScript, React, Next.js, Node.js, NoSQL
- **Cloud & Infrastructure:** GCP (Cloud SQL, Cloud Functions), Azure AI Foundry, AWS, Git, Supabase
- **Integrations:** WhatsApp Business Platform, Clerk, WorkOS

CERTIFICATIONS, TRAINING & LANGUAGES

- **Certifications:** Databricks Data Engineer Associate (2025) | Databricks Data Analyst Learning Path (2025)
- **Training:** Microsoft Learn — Develop AI agents on Azure; Develop generative AI applications on Azure (2026). Platzi/Udemy — 6 courses in ETL, data engineering, NLP, and BI.
- **Languages:** Spanish (native) | English (B2, upper intermediate)